## McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

# Dorm

## Location ID: B207

Three-story personnel dormitory housing up to 134 persons. The first floor includes 17 two-person rooms, a lounge, a public restroom, and a sauna. The second floor includes 24 two-person rooms and a small lounge. The third floor includes 26 two-person rooms.

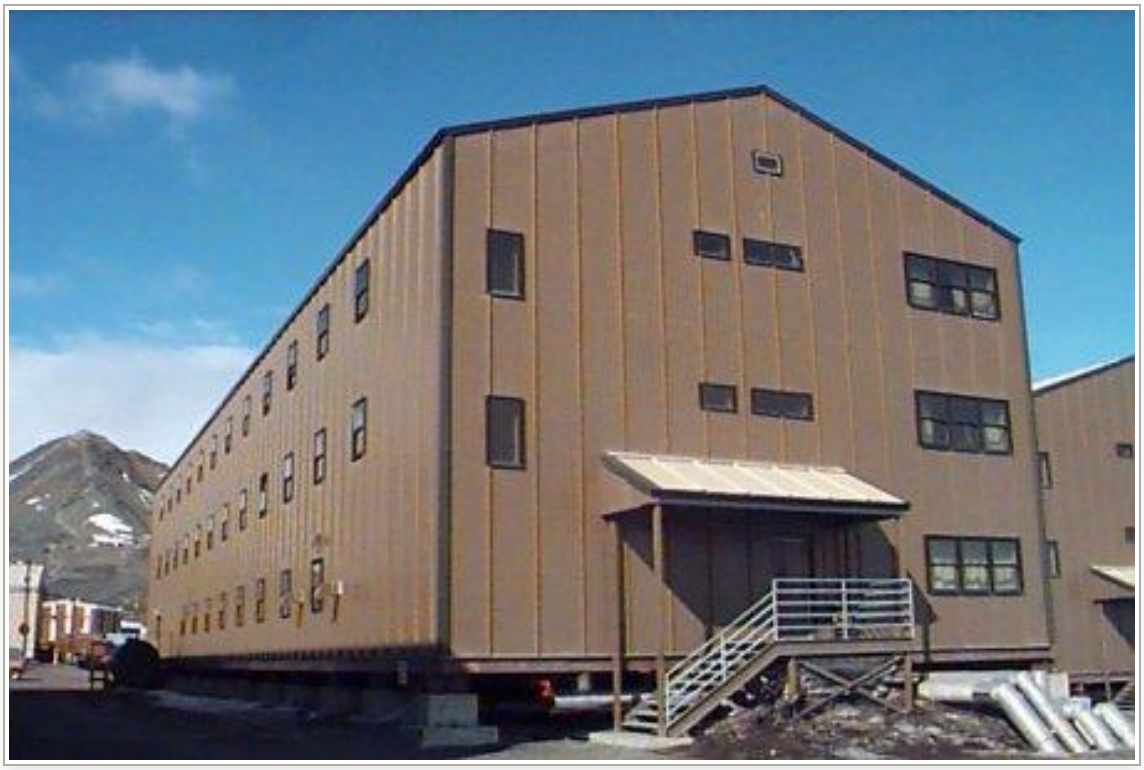


***Photo 1:*** *Dorm*

# General Structure Information

| | |
|---|---|
| **Usage** | Dormitory |
| **Occupancy Group** | R-1 Dormitories |
| **Maximum Occupancy** | 381 |
| **Construction Type** | Type II, 1-Hour, non-combustible; steel-framed structure with 2-1/2" thick foam-insulated metal siding and roofing, steel-framed<br>heavy timber first floor, steel-framed metal second and third floors. Steel-stud gypsum board interior partitions. |
| **Year Constructed** | 1986-87 |
| **Dimensions** | 48' X 168' |

**Floor Area**

| Floor | Area |
|---|---|
| Floor 1 | 8,064 sq ft |
| Floor 2 | 8,064 sq ft |
| Floor 3 | 8,064 sq ft |
| **Total** | 24,192 sq ft |

# HVAC System

| | |
|---|---|
| **Heating** | York-Shipley oil-fired glycol boiler (2,009,000 Btu/hr) |
| **Air Handling Units** | Two Trane, Climate Changer air handling units. They are located at each end of the building in the attic vestibule. They supply<br>fresh air only at 60 degrees F. |
| **Fuel Tank** | 2,500 gallon tank |

# Water and Sewer

| | |
|---|---|
| **Water Service** | 2 inch |
| **Sewer Service** | 4 inch |
| **Connections** | Sinks:<br>Toilets:<br>Urinals: 1<br>Showers:<br>Water Fountains: 1<br>Washing Machines: 6<br>Ice Machine: 2<br>Sauna: 1<br>100 lines - pending verification for all buildings |
| **Water Heater** | Glycol heat exchanger with 440 gallon hot water storage tank |

# Electrical Service

| | |
|---|---|
| **Service Feeder** | Feeder A |
| **Electrical Service** | 208/120V, 3PH, 400A main |
| **Phone and LAN** | 100 lines - pending verification for all buildings |

# Fire System

| | |
|---|---|
| **Fire Panel Type** | Pyrotronics System 3 control panel, King-Fisher KFRTI-WP-8 radio interface/transmitter panel, pull stations, alarm horns |
| **Suppression Type** | Pre-action sprinkler system connected to Firetrol pump and resevoir in Building 151 and equipped with Star, 6-inch, model D, electromagnetic<br>deluge valve with air pressure supervision. Deluge valve releases upon low air pressure and heat detector actuation. Firetrol pump starts<br>on water pressure loss in main trunk line to dorms. |

# Floorplans

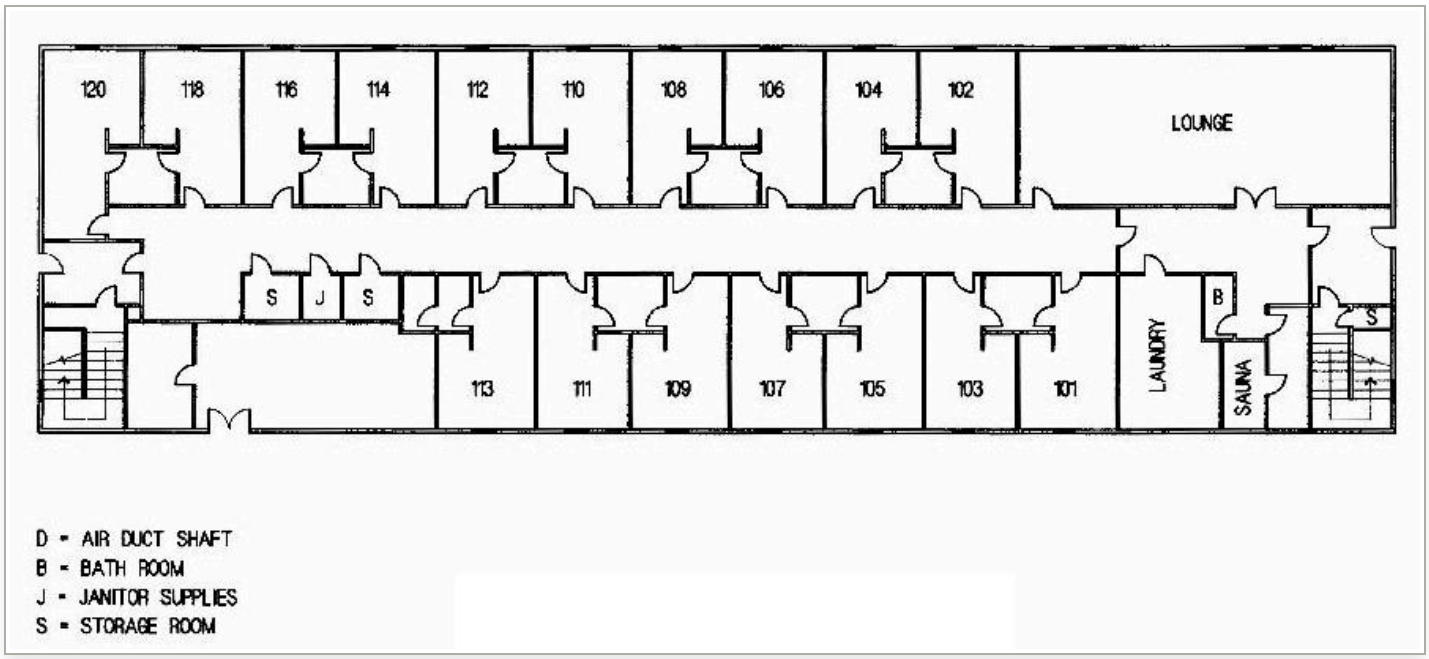



***Figure 1:*** *Dorm*

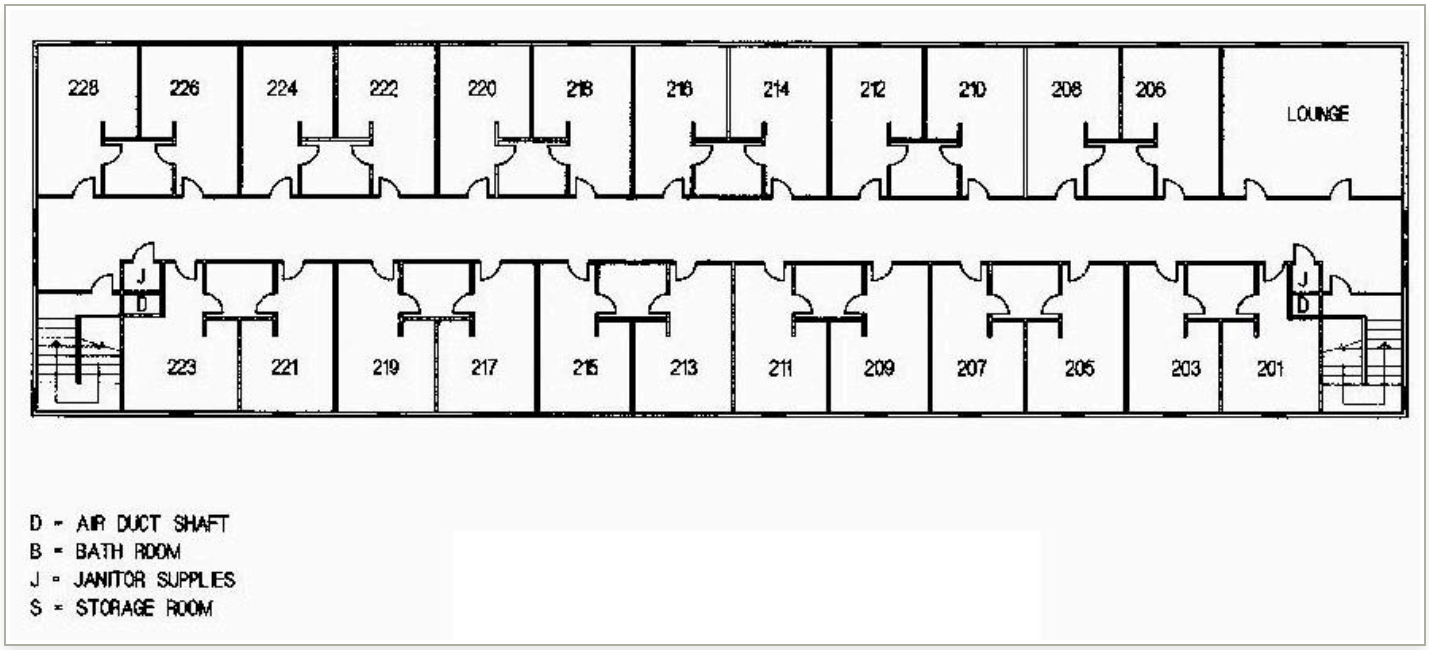



***Figure 2:*** *Dorm*

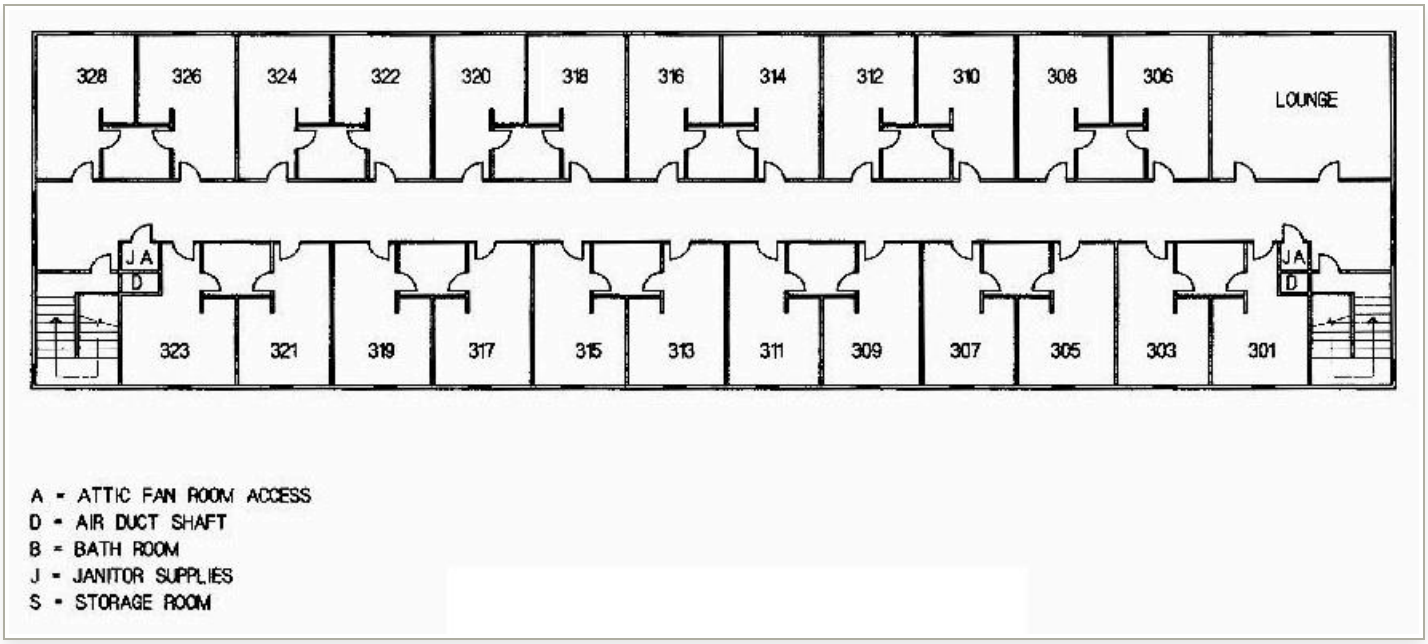



***Figure 3:*** *Dorm*